

- 9 (a) (i) fraction of the output (signal) is added to the input (signal)
out of phase by $180^\circ / \pi$ rad / to inverting input
- M1
A1 [2]
- (ii) e.g. reduces gain
increases bandwidth
greater stability
reduces distortion
(any two, 1 mark each)
- B2 [2]
- (b) (i) gain = $4.4 / 0.062$
= 71
- A1 [1]
- (ii) $71 = 1 + 120/R$
 $R = 1.7 \times 10^3 \Omega$
- C1
A1 [2]
- (c) for the amplifier not to saturate
maximum output is $(71 \times 95 \times 10^{-3})$ =) approximately 6.7 V
supply should be ± 9 V
- B1
M1
A1 [3]

- 10 (a) (i) strain gauge
- B1 [1]